Second-Order Conditions

ECON 441: Introduction to Mathematical Economics
Lecture 15

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Recap

For a continuously differentiable function $y = f(x)$:

- A critical point x_0 satisfies $f'(x_0) = 0$ (necessary for an interior extremum)
- First-derivative test: check whether $f'(x)$ changes sign at x_0

Today: a quicker test that uses the **second derivative**.

Second and Higher Derivatives

The derivative of $f'(x)$ is called the **second derivative**:

$$f''(x) = \frac{d^2y}{dx^2}$$

Similarly, we can obtain higher-order derivatives:

$$f'''(x),\ f^{(4)}(x),\ \ldots,\ f^{(n)}(x) \qquad \text{or} \qquad \frac{d^3y}{dx^3},\ \frac{d^4y}{dx^4},\ \ldots,\ \frac{d^ny}{dx^n}$$

Example: find $f''(x)$ for $f(x) = 2x^3 - 9x^2 + 12x$

What the Second Derivative Tells Us

With an infinitesimal increase in x from x_0:

- The **value** of the function increases if $f'(x_0) > 0$ and decreases if $f'(x_0) < 0$
- The **slope** of the function increases if $f''(x_0) > 0$ and decreases if $f''(x_0) < 0$

Second-Derivative Test

If $f'(x_0) = 0$, then $f(x_0)$ is

1. a **maximum** if $f''(x_0) < 0$
2. a **minimum** if $f''(x_0) > 0$

More convenient to use than the first-derivative test.

What if $f''(x_0) = 0$?

Example

Recall from Monday:

$$f(x) = 2x^3 - 9x^2 + 12x$$

Classify its critical points using the second-derivative test.

Example

$$f(x) = x^4 - 4x^3$$

Find the critical points. What does the second-derivative test say at each one?

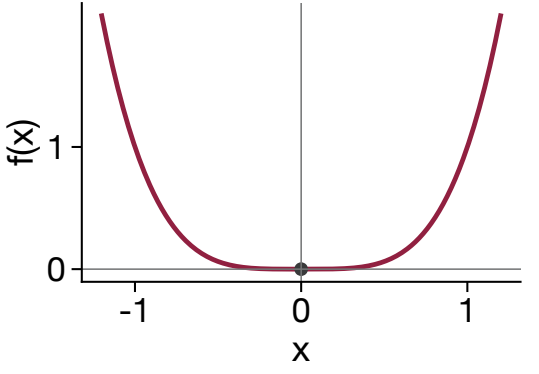
Necessary vs Sufficient Conditions

For a **maximum** at x_0:

- First-order necessary: $f'(x_0) = 0$
- Second-order necessary: $f''(x_0) \leq 0$
- Second-order sufficient: $f''(x_0) < 0$

For a **minimum**, reverse the inequalities. The second-order conditions apply only after the first-order condition is satisfied.

$y = x^4$



Homework Questions

- Exercise 9.4: 1 (b), (d), 2, 3

Textbook reference: 9.4